



Comparative evaluation of virtual reality setups for assessing cervical range of motion: a reliability study

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Abstract

Cervical pain is a highly prevalent musculoskeletal condition, affecting an estimated 222 million people globally. There is a clear need for novel technologies that enable more accessible and affordable care delivery for this population. Assessing the Cervical Range of Motion (CROM) of the patient is a necessary step before designing a treatment plan, and periodically during treatment to monitor patient progress. This study analyzes the use of three VR setups (HTC Vive Pro Eye, HP Reverb G2, and Meta Quest 2) for CROM assessment, representing high-end, mid-range, and low-end devices. 94 individual assessment sessions were conducted by the same rater using a custom-designed VR application. This application simulates a medical environment and guides users through the required neck movements while storing tracking data for later analysis. Intra-rater reliability, measured by the Intraclass Correlation Coefficient, ranged from good to excellent for all setups: 0.81–0.96 for HTC Vive Pro Eye, 0.92–0.93 for Meta Quest 2, and 0.89–0.94 for HP Reverb G2. The low-cost options presented slightly higher ICC than the high-end setup, although this difference was not statistically significant. The HMD physical design, weight distribution, and gripping mechanisms had a larger impact on the reliability than the tracking technology used. An anonymous survey showed high usability scores for all three devices. These findings support VR technology as a reliable and user-friendly tool for CROM assessment and show that low-cost devices can be used without compromising reliability or usability.

Keywords Virtual reality · Cervical range of motion · Rehabilitation · Physical therapy · Reliability · Neck pain

1 Introduction

Approximately 1.71 billion people worldwide live with musculoskeletal conditions, including low back pain, cervical pain, fractures, other injuries, osteoarthritis, amputations, and rheumatoid arthritis. These musculoskeletal conditions constitute the largest contribution to years lived with disability (YLDs) globally, accounting for approximately 149 million YLDs, or 17% all YLDs worldwide (World Health Organization 2022). Cervical pain is one of the most prevalent musculoskeletal conditions (Kazeminasab et al. 2022), with about 222 million cases worldwide, and it is responsible for 22 million YLDs. The social impact of these conditions extends beyond specific age groups, affecting both young and old individuals (Lewis et al. 2019). Hence, there is a need for new technologies that make the treatment of musculoskeletal conditions more efficient, accessible, and affordable (Chehade et al. 2020).

One of the most promising technologies to address this situation is Virtual Reality (VR). VR-based therapies have

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been shown to improve the outcomes of patients with musculoskeletal rehabilitation due to several factors, including increased patient motivation, greater personalization and adaptation of therapeutic goals, and a more immersive experience that enhances neuroplasticity (Brepohl and Leite 2023; Câmara et al. 2023; Huang et al. 2022). The use of VR has been associated with increased neuroplasticity in stroke patients (Huang et al. 2022) and has proven effective in adaptive cognitive training for addressing mental and behavioral disorders (Câmara et al. 2023). VR can be used as a tool for developing telerehabilitation solutions that could make rehabilitation treatments more accessible to patients and reduce the costs of healthcare services (Cerdeira et al. 2024; Moulaei et al. 2022). Telehealth and VR technologies have also been applied to chronic pain management, highlighting their impact on neuroplasticity (Cerdeira et al. 2024).

The first step to treat a musculoskeletal condition is to conduct an initial patient assessment, upon which the therapeutic goals are established and an appropriate action plan is designed (Cavedoni et al. 2022). Periodically, it is necessary to conduct new assessments to monitor the patient's progress and to adapt the rehabilitation goals accordingly. A fundamental metric for assessing the state of any joint is the range of motion; i.e., the number of degrees that the joint can move in its different planes of motion. In the case of the cervical range of motion (CROM), it is measured in three different planes: sagittal for flexion-extension (pitch), transversal for rotation (yaw), and frontal for lateral flexion (roll). It is important to note that compensatory movements, using partial trunk movements instead of solely moving the neck, can affect the CROM assessment results (van der Kruk et al. 2021). Such compensation occurs, consciously or unconsciously, when the typical muscle movements are altered due to pain or the ineffectiveness of the usual movement. Pain and negative thoughts contribute to the development of pain-related fears that, in the case of neck pain,

can cause the individual to avoid activities involving neck movement and can lead to compensatory movements (Kammler-Sücker et al. 2023).

Various devices are available to assess CROM, such as the CROM device (Audette et al. 2010; Wolan-Nieroda et al. 2020), inclinometers (Prushansky et al. 2010; Yoo et al. 2011), inertial sensors (Raya et al. 2018; Yoon et al. 2019) and goniometers (Luedtke et al. 2020; Williams et al. 2010). The goniometer is the most commonly used in clinical practice due to its portability and ease of use, but its measurements have a low reproducibility (Williams et al. 2010). Inertial sensors are cheap, provide more repeatable measurements, and can be integrated into clinical protocols, but their complexity renders them impractical for independent patient usage in telerehabilitation. The highest reproducibility assessment is provided by motion acquisition systems (Cánovas-Ambit et al. 2021; Niewiadomski et al. 2019) and by photogrammetry (Janjua et al. 2020). However, these solutions require expensive and complicated setups and the support of specialized technical personnel, which in practice limits their use in research studies.

VR setups can be used not only for the treatment of musculoskeletal conditions, but also for their assessment (Gumaa et al. 2021). The tracking system built into VR devices can provide three measures for positioning and three for orientation (Niehorster et al. 2017), for a total of six degrees of freedom (see Fig. 1). Thus, it enables the precise tracking of the position and orientation of the user's neck, through the Head Mounted Display (HMD). Also, by using additional external trackers, like the Vive Tracker, the movement of any other body part can also be captured. Therefore, one tracker placed in the trunk might help correct the compensatory movements of the patient, reducing their effect on the CROM.

Automatic CROM assessment can be highly valuable for designing personalized therapeutic interventions or adapting existing ones (Cerdeira et al. 2024; Snodgrass et al. 2014). A rehabilitation therapy utilizing a VR setup that performs this assessment can automatically adapt the exercises based on the patient's progress and capabilities, as determined by their CROM. Additionally, this assessment enables VR-based therapies to integrate home sessions using a VR setup with in-person sessions guided by healthcare professionals. This hybrid approach not only increases patient engagement through interactive and gamified exercises (González-González et al. 2019) but also provides therapists with objective information to adjust and optimize therapy according to patient progress.

In the first work in the literature to use VR to measure neck movement (Xu et al. 2015), the neck position was obtained using an Oculus Rift and compared with measurements taken by a motion capture system, obtaining errors

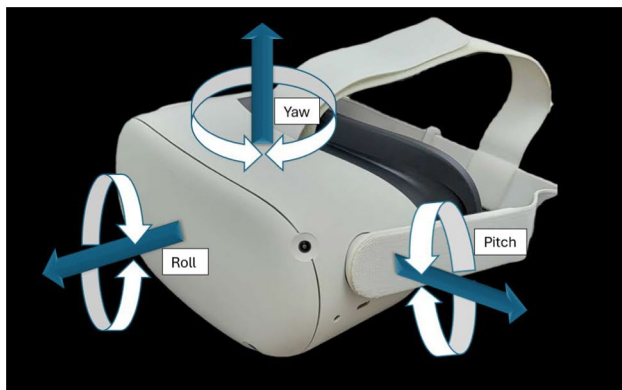


Fig. 1 VR six degrees of freedom, encompassing three translation movements along the X, Y, and Z axes, as well as three rotation angles around each axis (pitch, yaw, and roll)

in the measurements of up to 10° . Based on these errors, the authors concluded that the VR device was only suitable for obtaining approximate estimates of neck position. An important limitation of this study is that only moderate-range movements were performed, not reaching the maximum range of the joint. To assess the CROM, it is necessary to perform movements until the maximum range of the joint is reached. In (Kiper et al. 2020) the CROM assessment obtained using a non-immersive system based on an avatar displayed on a computer screen was compared with that obtained using an Oculus Rift, concluding that the VR system provided more accurate measurements. However, this work did not carry out a reliability study of the measurements obtained. Reliability in the context of a joint's range of motion assessment refers to the consistency and repeatability of the measurements obtained. Good reliability ensures that the measurements can be used to track a patient's progress over time (Rojo et al. 2022).

To the best of our knowledge, the first study in the literature on the reliability of VR as a CROM assessment tool was (Santos-Paz et al. 2022). In this work, a high-end VR setup (HTC Vive Pro Eye) was used, showing that it was possible to achieve good to excellent reliability (interclass correlation coefficients (ICC) (Trevethan 2017) values between 0.86 and 0.96). In this study, a high-end VR device was chosen under the hypothesis that if a reliable CROM assessment could not be achieved with a high-end VR setup, it would not be possible with a lower-end device either. The cost of the VR set-up used in the study and the need for a computer to run the VR application make a VR-based rehabilitation solution using this device expensive. Hence the interest in studying whether a reliable CROM assessment can also be achieved with lower-cost VR setups.

The main objective of this work is to compare three different VR setups by measuring their intra-rater reliability (Trevethan 2017). The devices tested were chosen as representatives of the multitude of commercially available solutions, belonging to three different categories: high-end setups with tracking using external base stations and with the HMD tethered to a PC, mid-range setups using inside-out tracking with the HMD tethered to a PC, and low-end setups using inside-out tracking and a standalone HMD.

2 Materials and methods

All VR devices tested can track both the translation and rotation of the HMD and two controllers within the virtual environment. The rotation of the HMD provides the data needed for the calculation of the CROM. The movement of flexion-extension corresponds to the sagittal plane (pitch), the movement of right-left rotation to the transverse plane

(yaw), and the movement of right-left lateral flexion to the frontal plane (roll) (Kauther et al. 2012) (see Fig. 1).

There are two main differences between the three VR setups tested. The first one lies in the tracking system used. The high-end setup uses two base stations, placed in fixed positions around the tracked space, opposite to each other. They emit infrared laser arrays that sweep the area horizontally and vertically, hitting sensors placed on the surface of the HMD and the controllers. These sensors are photodiodes that register temporal variations in the laser impacts. By analyzing the time differences between hits on different photodiodes, precise tracking of the devices' three-dimensional position and orientation is achieved.

The mid-range and low-end setups share the same base technology for tracking, as both utilize an inside-out approach. In it, integrated cameras placed in the HMD capture images of the environment and the controllers' movements (or user's hands, if hand-tracking is enabled), allowing the device to track their position and movements in space without external sensors.

All setups also have inertial measurement units (IMUs) placed inside the HMD and the controllers, and perform a sensor fusion between these data and the information retrieved by their other tracking technology.

The second difference lies in the need for, or lack of, a graphically powerful PC that renders the images shown in the HMD and processes the tracking data. In both the high-end and mid-range setups, the HMD needs such a PC. In the low-end setup, the HMD itself contains all the processing elements needed. Thus this device can work untethered from a computer. This makes the low-end setup stand out for its versatility and portability.

The high-end setup features an HTC Vive Pro Eye HMD, two wireless HTC Vive Controllers, and two Lighthouse v2 base stations. The HMD has a resolution of $1,440 \times 1,600$ pixels per eye, a refresh rate of 90 Hz, and a field of view of 110° (see Table 1). The computer used was an HP OMEN Intel Core i7-10750 H with 16GB of RAM and an 8GB NVIDIA GeForce RTX 2070 Super GPU, running the 64-bit version of Windows 10 Pro, SteamVR 1.20.10, and the SteamVR version of the CROM assessment application, that shall be described in the next section.

This setup supports the addition of one or more Vive Tracker motion trackers that can be positioned on arbitrary body parts or objects, enabling their representation within the VR experience. Taking advantage of this feature, a tracker was placed on the users' backs to measure compensatory movements of the trunk.

The mid-range setup featured an HP Reverb G2 HMD and two HP wireless controllers. The HMD has a resolution of $2,160 \times 2,160$ pixels per eye, a refresh rate of 90 Hz, and a field of view of 114° (see Table 1). The computer

Table 1 Technical specifications of the VR head-mounted displays used in the study

	HTC Vive Pro Eye	HP Reverb G2	Meta Quest 2
Device Type	PC VR	PC VR	Standalone VR
Weight (g)	800	550	503
Refresh rate (Hz)	90	90	72, 90, 120
Horizontal FOV (10°)	98	98	97
Vertical FOV (10°)	98	90	93
Diagonal FOV (10°)	110	114	98
Display	2 x AMOLED	2 x LCD	1x Fast Switch LCD
Resolution (per-eye)	1440 10° 1600	2160 × 2160	1832 × 1920
Tracking method	6 DoF Marker-based, 2 x Vive Base Station 2.0	6 DoF Inside-out via 4 integrated cameras	6 DoF Inside-out via 4 integrated cameras
Controller type	Vive Pro controller	WMR controller	Oculus Touch v2
Special features	Eye Tracking	—	Hand Tracking

used was an HP OMEN Intel Core i7-10750 H with 16GB of RAM and an 8GB NVIDIA GeForce RTX 2070 Super GPU, running the 64-bit version of Windows 10 Pro, SteamVR 1.20.10, and the SteamVR version of the CROM assessment application.

The low-end setup featured a Meta Quest 2 HMD and two Meta Touch 2 controllers. The HMD has a resolution of 1832 x 1920 pixels per eye, a field of view of 98°, and was running under a refresh rate of 90 Hz (see Table 1). This setup used the Oculus OS version of the CROM assessment application.

2.1 Participants

The protocol for this study was approved by the Research Ethics Committee of the University San Pablo-CEU, approval code: 549/21/48. All experimental procedures were carried out in accordance with the relevant guidelines and regulations. Prior to participation, all participants provided written informed consent after receiving a complete verbal description of the procedures and the purpose of the study.

Participants were selected from students and faculty members of the University San Pablo-CEU. Men and women of all races, aged 18 to 65 years, who had not experienced cervical joint pain in the last month and had not received prior treatment for neck pain were included. In order to meet the requirements for conducting a reliability study (Koo and

Li 2016), the aim was to measure at least 30 subjects for each VR setup. A total of 94 measurement sessions were conducted across the three VR setups. These 94 sessions involved 85 unique participants; of these 5 were assessed using two setups, and 2 were assessed using all three, with the rest being evaluated with a single VR setup. To minimize potential learning or fatigue effects, these seven participants completed their assessments on different days.

Subjects were divided as follows: for Meta Quest 2 there were 30 subjects: 17 men and 13 women with an average age of 28.90 ± 11.84 years. Regarding HP Reverb G2, 30 subjects participated: 13 men and 17 women with an average age of 27.02 ± 8.08 years. For HTC Vive Pro Eye, a total of 34 subjects participated, comprising 16 men and 18 women with an average age of 25.76 ± 7.86 years. This last set of subjects also had a Vive Tracker set on their back.

This resulted in a total of 42 men with an average age of 29 ± 11 years and 43 women with an average age of 25 ± 6 years. Table 2 presents the anthropometric information of the subjects.

2.2 VR CROM assessment application

The virtual environment simulates a medical/physiotherapy office, designed sparsely to avoid distractions during the assessment procedure. The virtual room contains a desk, two chairs with a human figure seated in one of them, another chair where the subject's avatar is seated, and some medical furniture placed further away.

The CROM assessment process requires the user to perform three movements: flexion-extension, rotation and lateral flexion. To instruct the subject to perform these movements, there is a figure positioned in front of the user in the virtual space that demonstrates them, accompanied by an arrow (see Fig. 2 bottom-left corner). Audio instructions played through the HMD's integrated headphones also guide the user through the process and specify the movements to be performed. To obtain greater reliability in CROM measurements, the application instructs the subject to perform each movement three times, for a total of nine movements; the final CROM measurement in the corresponding plane is the average of the three repetitions (Raya et al. 2018). As users perform the movement, a large black number in the center of the field of view shows the current repetition number (see Fig. 2 bottom-right corner).

Two versions of the CROM application have been developed to accommodate different VR platforms. One was designed for devices compatible with SteamVR (see Fig. 2 top-left corner), while the other was created for devices running on the Oculus OS operating system (see Fig. 2 top-right corner).

Table 2 Anthropometric information of the participants in the evaluation of the different VR setups

Setup	Gender	Subjects	Age	Weight	Height	BMI
HTC Vive Pro Eye	Male	16	26.31 ± 9.49	79.28 ± 13.74	1.79 ± 0.06	24.69 ± 3.54
	Female	18	25.28 ± 6.31	56.98 ± 7.61	1.65 ± 0.06	20.98 ± 2.71
	Total	34	25.76 ± 7.86	67.47 ± 15.60	1.72 ± 0.09	22.73 ± 3.61
Meta Quest 2	Male	17	33.35 ± 13.97	80.00 ± 16.54	1.76 ± 0.07	25.81 ± 4.43
	Female	13	23.08 ± 3.73	58.00 ± 8.78	1.63 ± 0.06	21.85 ± 2.76
	Total	30	28.90 ± 11.84	70.47 ± 17.49	1.70 ± 0.09	24.10 ± 4.24
HP Reverb G2	Male	13	27.85 ± 8.80	80.92 ± 22.76	1.78 ± 0.08	25.44 ± 5.60
	Female	17	26.71 ± 7.73	58.59 ± 8.20	1.64 ± 0.07	21.83 ± 2.44
	Total	30	27.20 ± 8.08	68.27 ± 19.45	1.70 ± 0.10	23.39 ± 4.42
Overall Population	Total Male	42	29.10 ± 11.72	77.63 ± 14.46	1.77 ± 0.07	24.82 ± 3.94
	Total Female	43	25.19 ± 6.60	58.11 ± 8.25	1.64 ± 0.07	21.54 ± 2.61
	TOTAL	85	27.22 ± 9.36	68.68 ± 17.36	1.71 ± 0.10	23.38 ± 4.08

Of the 34 participants that used the HTC Vive Pro Eye, 29 were only assessed with this device, and 5 were also assessed with other HMDs. Of the 30 that used the Meta Quest 2, 24 were only assessed with this device, and 6 were assessed with another HMD. Of the 30 that used the HP Reverb G2, 25 were only assessed with this device, and 5 were also assessed with other HMDs. The Overall Population values reflect unique participants, ensuring that individuals assessed with multiple HMDs are counted only once. Age, weight, height, and body mass index (BMI) values are presented as mean ± standard deviation

The first version is an executable file designed for use with SteamVR, compatible with the mid-range and high-end setups. In this version, the application is made up of two interfaces: the VR experience seen by the user through the HMD and an interface displayed on the computer monitor for the evaluator (see Fig. 2). The evaluator interface allows the entry of a unique number associated with each subject, used to identify all evaluation sessions of each participant. Additionally, it has control buttons to initiate the evaluation and exit the application.

The second version is an APK file designed specifically for devices running on the Oculus OS, such as the low-end HMD. This portable version allows the users to input their own identification number and select the usage mode of the application directly through a panel shown inside the virtual environment. Two buttons can be used to increment and decrement the user ID. The assessment itself is carried out the same way as in the first version of the application, ensuring consistency in both cases.

Both applications offer two assessment modes. The first mode provides detailed instructions to guide users through the movements. Since the average user is not familiar with CROM assessment, it is important to ensure their complete understanding of the required procedures and movements. Therefore, these detailed explanations are needed before performing the movements for the first time. Additionally, a warm-up session is conducted before starting the assessment to properly prepare the user's body. The warm-up consists of performing all the movements that would be part of the assessment once. This initial data is discarded, and the subject has to repeat all the movements to perform the CROM assessment. The warm-up procedure helps prepare muscles and joints prior to movement. It increases blood

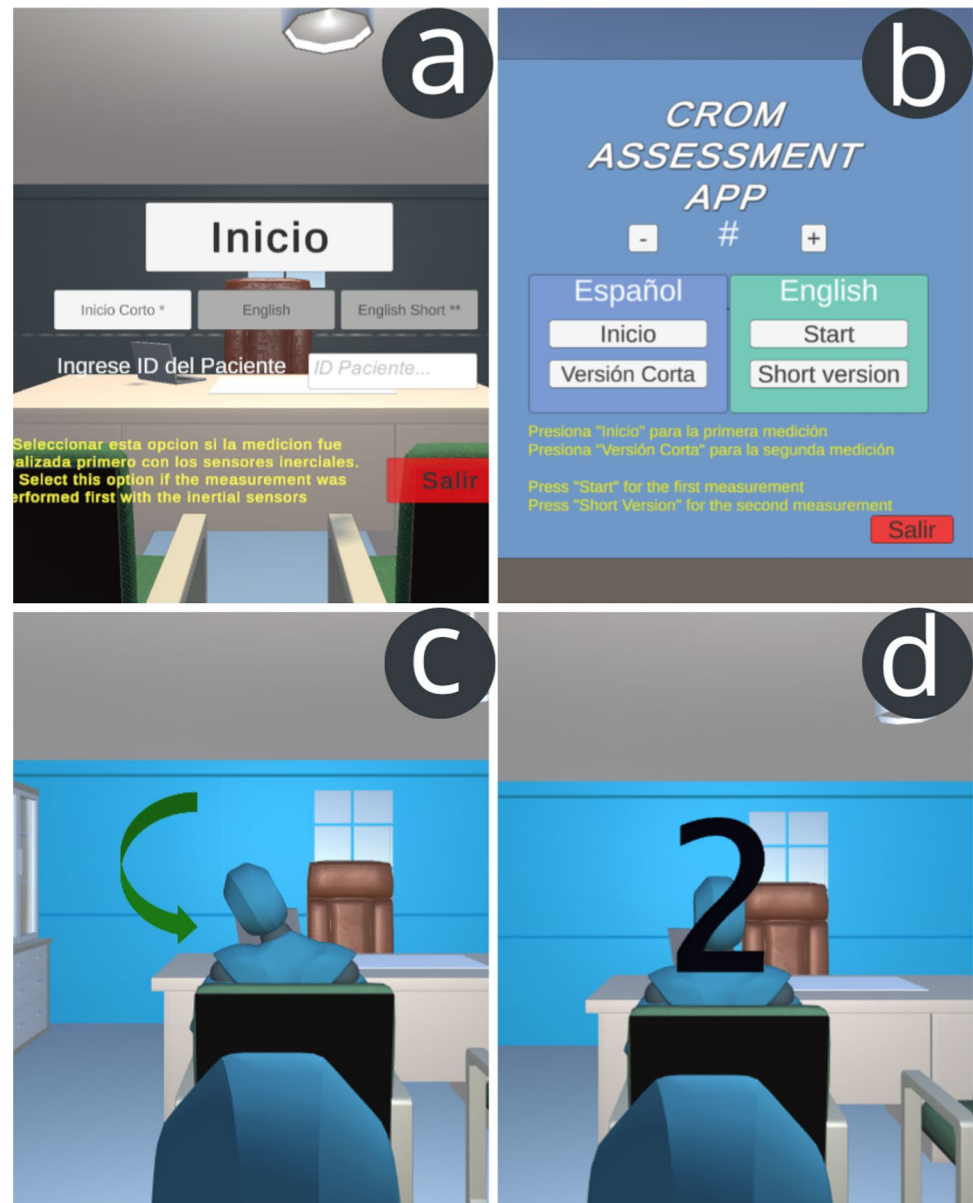
flow and joint flexibility, allowing participants to achieve a consistent maximum range of motion (ROM) (Knudson 2018). When performing dynamic movements, warming up activates the muscles involved in cervical movement, reducing variability between tests and ensuring more reliable measurements (Afonso et al. 2024). This helps minimize the effects of muscle stiffness on CROM results.

The second mode offers a simplified set of instructions and skips the warm-up session. This mode is designed for users who have just completed the first assessment and are therefore familiar with the movements and procedure. The fact that they have just completed an assessment also eliminates the need for a warm-up. By omitting the warm-up and providing concise instructions, this mode aims to simplify the user experience, as less guidance will be needed during the second assessment.

For the SteamVR application, the evaluator selects a different button on the PC screen interface depending on whether it is the first or the second assessment. For the Oculus OS application, the subjects themselves select between the two operating modes (see Fig. 2 top-right corner).

The SteamVR application was developed using Unity version 2019.4.19f1 with the OpenVR XR plugin version 1.1.4. The Oculus OS application was developed using Unity version 2021.2.12f1 with the XR Interaction Toolkit package version 2.0.1 and the OpenXR plugin version 1.3.1. The code that records the tracking data is the same for both versions of the application; it is placed in a script attached to the main camera, in the FixedUpdate() function of the game loop, to remove any frame-rate dependency. In the high-end and mid-range setups, this function is called with a frequency of 90 Hz, while in the low-end setup it is called with a frequency of 50 Hz. This last frequency was

Fig. 2 Graphical interface of the CROM assessment application. **a** Start menu of the application for tethered devices, which runs on the computer and is visible to the evaluator. This interface contains four buttons to initiate the long and short versions of the assessment in both Spanish and English, as well as a field to enter the user ID. **b** Start menu of the application that runs on the standalone device, with the same structure of four buttons and user ID input field. **c** Example of the left lateral flexion movement instruction. **d** Movement repetition indicator



chosen with the limitations of a standalone device in mind; its reduced computing capabilities limit how many times a second the physics engine calculations (which are tied to the frequency chosen for `FixedUpdate()`) can be performed. On top of that, the Meta Quest 2 tracking system samples at the frequency of the mains electricity of the country it is running in, which in our case is also 50 Hz. This synchronization prevents tracking inaccuracies caused by AC-powered lights flickering at a different frequency than the tracking system sampling. CROM assessment focuses on the maximum range, rather than the exact position during every part of the motion. The maximum range is reached when the movement speed in one direction decreases until motion stops, and then starts increasing in the opposite direction. Hence the average speed around maximum CROM is the

lowest during the movement, and the 20 ms sampling period has a negligible impact on the measurements.

Tracking data was stored in CSV files, with each sample containing positional and orientation information for the HMD and, when available, for the external tracker. Positional data was recorded in Cartesian coordinates, while orientation was represented using both quaternions and Euler angles.

It is important to note that in the case of the high-end setup, the data from the tracker located in the back of the subject are also recorded. These data shall be used to correct compensatory movements.

2.3 Protocol

The same procedure was employed to carry out the assessments with the three VR setups. Prior to starting the measurements, the physical area was prepared by delineating and framing a 2x2 m space on the floor with adhesive tape, as can be seen in Fig. 3. This ensured that the virtual room used in all measurements corresponded to the same physical space. Although this process is essential only for measurements with the high-end setup (which uses base stations), the same area configuration was maintained for all assessments to ensure consistency across different setups. Additionally, a chair was placed at the center of this area, with its position marked on the floor to, again, ensure consistency.

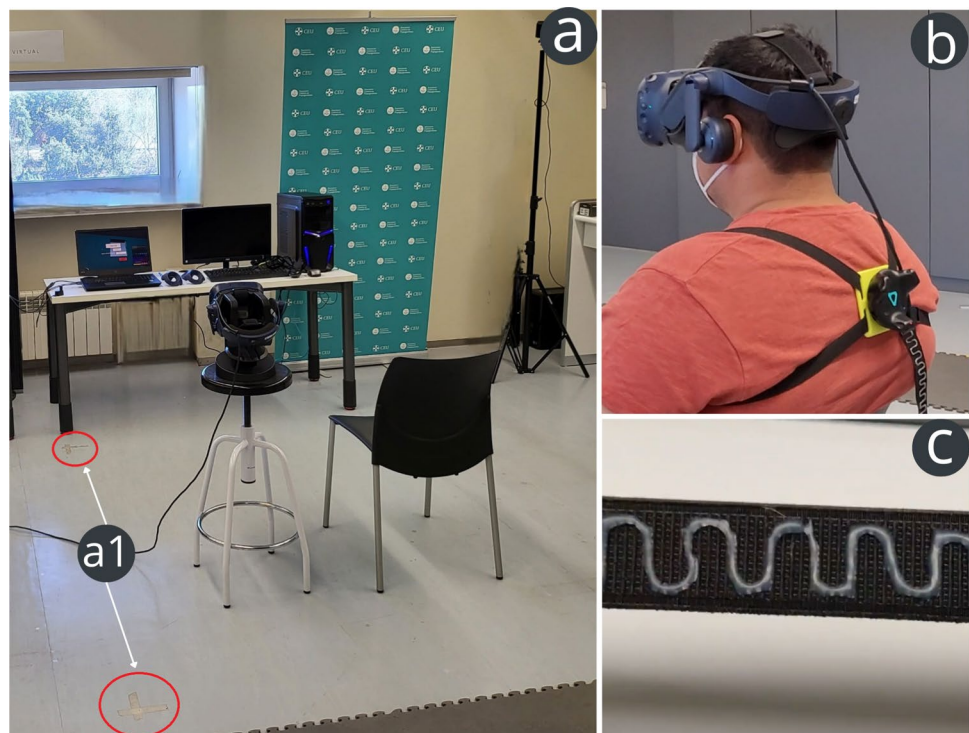
To assess trunk compensations with the high-end setup, a tracker was employed to monitor the movement of the subjects' back, as shown in the top-right corner of Fig. 3. The tracker, featuring a blue triangle at its center, was positioned over the fourth thoracic spine (T4), a location chosen for its demonstrated reliability in previous research (Raya et al. 2018; Tsang et al. 2013), and to minimize interference with the HMD during cervical extension movements. The attachment of the tracker utilized a custom X-shaped harness, crafted from a 2-meter long, 20-mm wide elastic band equipped with flexible silicone stripes to enhance grip. These silicone stripes, positioned on the side in contact with the subject, aimed to prevent tracker displacement, as can be seen in the bottom-right corner of Fig. 3. The harness incorporated two handles, each positioned at shoulder height on opposite arms for ease of use.

During the assessment sessions, a single evaluator provided support to the subjects in using the VR equipment. This evaluator has experience in using VR applications and is familiar with the CROM assessment application. For the high-end and mid-range setups, the evaluator controlled the CROM assessment application from the computer. For the low-end setup, the evaluator explained the operation of the CROM assessment application and assisted in placing the HMD, while the subjects initiated the measurement from an interface visible only from the HMD screen, inside the virtual environment.

The measurement protocol involved conducting two CROM assessments for each subject with one of the setups. Before commencing the first evaluation, the evaluator instructed the subject on the operation of the VR application and the correct use of the controllers. Specific instructions on how to perform the movements for the CROM evaluation were provided exclusively through the VR experience, while the evaluator only assisted with the correct placement of the HMD and the operation of the VR setup.

During the evaluations, subjects were free to use either hand to control the application with the VR controllers provided. The movement sequence for each subject was randomly generated and was composed of one flexion-extension, one rotation, and one lateral flexion, with no repetitions within the same sequence. Randomly generating the movement sequence for each subject is necessary to ensure unbiased evaluation and eliminate potential learning or habituation effects.

Fig. 3 Experimental setup of the CROM assessment. **a** Assessment area, with white arrows and the indicator *a1* highlighting the tape that delimited the 2×2 m physical measurement space, ensuring that all measurements were conducted within the same area. The HTC Vive Pro Eye setup, including the base stations, controllers, and the evaluator's computer, are also visible. **b** Location of the tracker on the subject's back. **c** Tracker's securing strap with soft silicone bands to prevent slipping or movement during measurements



The first assessment began with the triggering of a brief warm-up and detailed movement instructions on the HMD screen. After completing the first assessment, the HMD was removed, and a brief pause was made before starting the second. The process of the second assessment followed a similar procedure but without warm-up and detailed instructions. During the whole process the evaluator supervised without intervening, providing assistance only if the subject needed support with the usage of the VR setup. The process of placing and adjusting the HMD can impact measurement accuracy. Therefore, as part of the protocol to simulate a complete evaluation, the HMD is carefully placed and adjusted at the beginning of each assessment and removed upon completion. This also allows the subject a brief rest interval before starting the next assessment.

2.4 Extraction of CROM data and statistical analysis

Matlab R2020a was used to perform the CROM calculation, processing the data of each subject individually. The CROM assessment VR application recorded the pitch, yaw, and roll data, corresponding to flexion-extension, rotation, and lateral flexion, respectively, with angles ranging from 0 to 360 degrees. These angles were mapped to the interval (-180 to 180) using a wrapping function. In this interval, 0 degrees represented the neutral position of the head before movement initiation. Positive angles indicated flexion or right rotation, while negative angles indicated extension or left rotation. For lateral flexion, positive angles represented left lateral flexion and negative angles represented right lateral flexion. A correction was applied to compensate for minor misalignments between the subject's head and the HMD's position in the neutral position. This is achieved by subtracting the initial orientation of the HMD from the instant before the start of movements, thus centering the angle of movement around 0.

Only in the case of the high-end setup, the removal of compensatory movements was performed by subtracting the angle obtained by the tracker placed in the subject's back from the angle obtained by the HMD in each time point. This way, the movement of the head is isolated from other body parts.

Using a Matlab application designed for this purpose, the angles recorded by the HMD were graphically plotted against time, allowing for visual identification of the three repetitions of each movement. Subsequently, time positions before and after each repetition are selected with a mouse. Using these positions, the Matlab application automatically identifies the three repetitions of the movement in the selected interval and extracts the maximum and minimum angles for each of the three movements (see Fig. 4). Finally, the average CROM value for each axis of movement based

on these three repetitions is calculated. If data on the user's back position are available (high-end setup), the calculation is performed with and without correction of compensatory movements.

IBM SPSS Statistics version 29 was used for statistical analysis. The normality of data distribution was assessed using the Shapiro-Wilk test, and descriptive statistics including mean and standard deviation were calculated. Differences in anthropometric information between the groups were assessed by conducting an ANOVA for age and BMI, and a chi-squared test for gender. Additionally, Intraclass ICC was used to assess the reliability of measurements performed by the same evaluator on different occasions. This means that the ICC provides information about the consistency of CROM measurements made by the same evaluator at different times. The ICC (2, k) model was used for intra-rater analysis (Koo and Li 2016). ICC values ranging between 0 and 0.5 are considered unreliable, while values between 0.5 and 0.75 indicate moderate reliability. ICC values between 0.75 and 0.9 are deemed good, and values above 0.90 are considered excellent (Koo and Li 2016).

In addition to the ICC, the Standard Error of Measurement (SEM) and Smallest Detectable Difference (SDD) were calculated (Knols et al. 2009). The SEM is a statistical measure that quantifies the amount of random error or variability in a measurement. It represents the average distance between an individual's true measure and the observed measure. It can be calculated using the formula:

$$SEM = \sqrt{\text{Mean Squared Error (} MS_E \text{)}} \quad (1)$$

The SDD reflects the minimal change in a variable between two measurements that can be reliably interpreted as a true difference (Stoustrup et al. 2013), which is also expressed as a percentage (SDD%). It represents the precision with which an assessment method can distinguish between true changes in a patient's condition. The SDD was calculated at a 95% confidence level using the formulas:

$$SDD = 1.96 \times SEM \times \sqrt{2} \quad (2)$$

$$SDD\% = \frac{SDD}{\text{Grand Mean}} \times 100 \quad (3)$$

To assess statistical differences between the ICC values for the different setups we have followed the methodology proposed by (Bhandary and Fujiwara 2013).

To assess the usability of each VR setup as a tool for CROM assessment, at the end of the CROM evaluation measurements session the subjects were asked to anonymously complete the System Usability Scale (SUS) questionnaire. The Spanish version of the SUS was used (Sevilla-Gonzalez

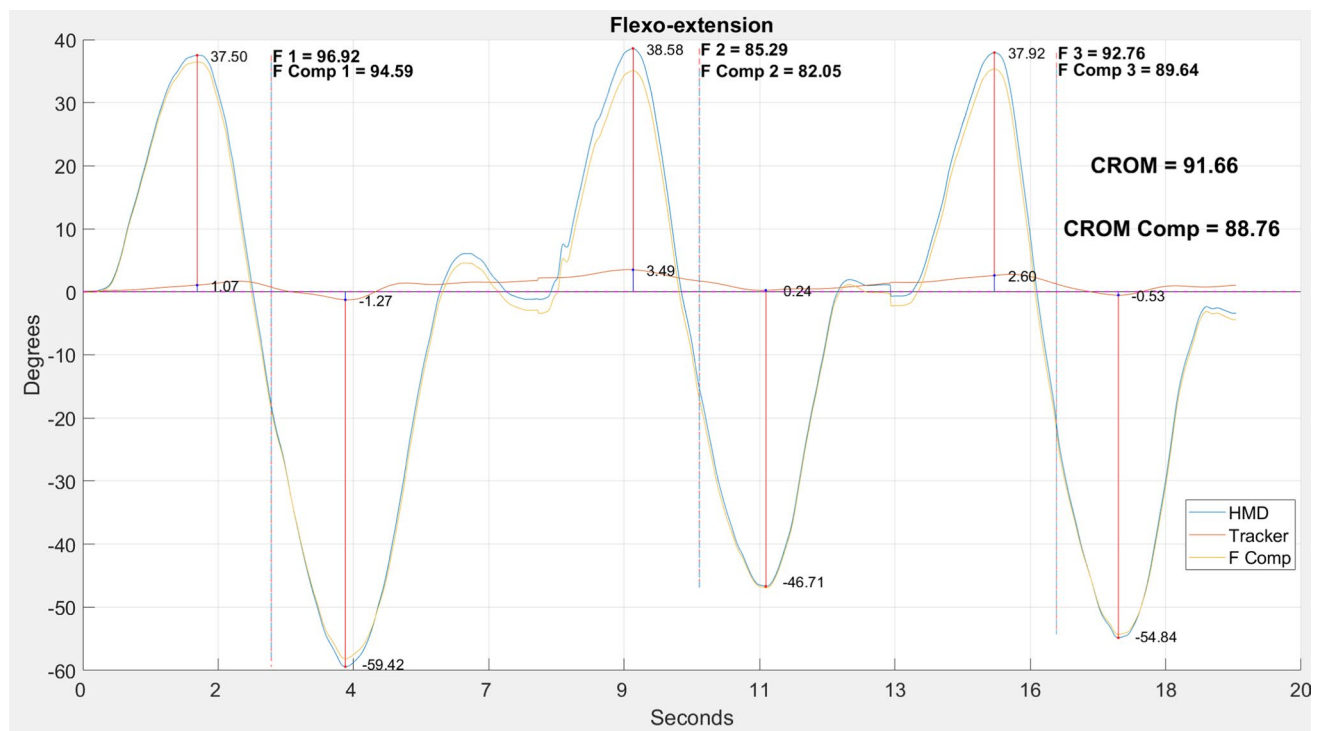


Fig. 4 Screenshot of the MATLAB application used to extract CROM. The image displays three repetitions of a flexion-extension movement. The blue line represents the head movement captured by the HTC Vive Pro Eye's HMD. The red line shows the compensatory back movement captured by the tracker. Finally, the orange line represents the

movement resulting from subtracting the back's compensation from the head movement. The maximum and minimum values for each repetition are also displayed. Additionally, the CROM and corrected CROM, calculated as the average of the three movements, are shown

et al. 2020). The SUS is a user-friendly and efficient tool for both system users and researchers that consists of 10 questions designed for users to answer after interacting with a system (Bangor et al. 2008). The questions alternate between positive and negative statements to mitigate respondent bias. Responses are scored from 1 to 5, where 5 represents "Strongly agree" and 1 indicates "Strongly disagree". Positive statements are scored by subtracting one point, while negative statements are scored by subtracting 5 minus the user answer. The total score ranges from 0 to 40, which when multiplied by 2.5, produces a scale from 0 to 100 (Brooke 1996). A score between 68 and 84 indicates good usability, while a score above 85 signifies excellent usability (Sevilla-Gonzalez et al. 2020).

3 Results

Table 3 presents both mean and maximum values (with corresponding SDs) of the CROM measurements in each of the planes of movement obtained for each VR setup. The Shapiro-Wilk analysis revealed that the distributions of CROM movements were normal in all cases (p -value > 0.05). An ANOVA test was used to compare age and BMI across the three experimental groups using different VR setups, and

a chi-square test was used to compare gender. These tests yielded p -values of 0.41, 0.41 and 0.57, respectively, indicating no statistically significant differences in age, BMI, or gender between the groups.

ANOVA tests showed no statistically significant differences in CROM measurements across the devices for any of the movements. Specifically, the p -values for flexion-extension, rotation, and lateral flexion were 0.57, 0.48, and 0.62, respectively, indicating that the devices did not differ in CROM measurements. These results suggest that the populations are comparable in terms of the measurements obtained across the different HMDs.

To assess statistical differences between the ICC values for the different setups, we calculated z -scores associated with differences in ICC for each pair of VR setups for each of the three planes of movement (Bhandary and Fujiwara 2013). For all comparisons, the uncorrected p -values were > 0.44 , indicating no statistically significant differences between the setups. Therefore, we conclude that the performance of the various HMDs in terms of the reliability (ICC) does not show statistically significant differences.

The ranges for the flexion-extension movement varied from 113° (HTC Vive Pro Eye) to 117° for HP Reverb G2 and Meta Quest 2. For lateral flexion, the ranges varied from 92° to 95° , and for rotation, they remained between 139°

Table 3 Maximum and mean cervical range of motion (CROM) for the different VR setups

Parameters	Maximum	Mean
<i>HTC Vive Pro Eye</i>		
Flexion-extension	118.68 ± 17.37	113.74 ± 17.50
Lateral Flexion	95.61 ± 16.16	91.75 ± 15.38
Rotation	143.14 ± 16.40	139.15 ± 16.18
<i>HTC Vive Pro Eye Removing Compensation</i>		
Flexion-extension	111.88 ± 14.79	107.38 ± 15.35
Lateral Flexion	85.09 ± 12.64	82.20 ± 12.68
Rotation	138.70 ± 14.13	135.19 ± 14.22
<i>HP Reverb G2</i>		
Flexion-extension	122.31 ± 17.12	117.52 ± 16.92
Lateral Flexion	99.45 ± 17.40	95.55 ± 16.77
Rotation	148.25 ± 17.71	143.62 ± 17.22
<i>Meta Quest 2</i>		
Flexion-extension	121.96 ± 19.02	117.39 ± 18.53
Lateral Flexion	97.50 ± 17.53	93.53 ± 15.66
Rotation	148.08 ± 18.82	143.25 ± 18.13

CROM measurements across the different movement planes, obtained with four VR devices: HTC Vive Pro Eye, HTC Vive Pro Eye with compensation removed (using a Vive Tracker placed on the back), HP Reverb G2, and Meta Quest 2. Values are presented as Maximum ± SD and Mean ± SD

Table 4 Intraclass correlation coefficient (ICC), 95% confidence intervals (CI) for ICC, standard error of measurement (SEM), and smallest detectable difference (SDD) for CROM assessment using different VR devices: HTC Vive Pro Eye, HTC Vive Pro Eye removing trunk compensation movements, HP Reverb G2, and Meta Quest 2

Device	Parameter	ICC	Confidence Interval (CI)		SEM	SDD	SDD%
			Lower	Upper			
HTC Vive Pro Eye	Flexion-extension	0.86	0.67	0.94	7.96	22.07	19.40
	Lateral Flexion	0.94	0.88	0.97	5.11	14.18	15.45
	Rotation	0.93	0.85	0.97	5.52	15.31	11.00
HTC Vive Pro Eye Removing Compensation	Flexion-extension	0.81	0.62	0.90	8.50	23.56	21.94
	Lateral Flexion	0.96	0.91	0.98	3.70	10.26	12.49
	Rotation	0.93	0.86	0.97	4.97	13.77	10.18
HP Reverb G2	Flexion-extension	0.89	0.73	0.95	6.95	19.26	16.39
	Lateral Flexion	0.94	0.88	0.97	5.57	15.43	16.15
	Rotation	0.84	0.66	0.92	9.12	25.27	17.59
Meta Quest 2	Flexion-extension	0.92	0.78	0.97	6.52	18.07	15.39
	Lateral Flexion	0.93	0.84	0.97	5.29	14.66	15.68
	Rotation	0.93	0.86	0.97	6.68	18.53	12.93

and 143°, with HTC Vive Pro Eye being the lowest in all cases. The correction of back compensatory movements in the case of the HTC Vive Pro had an average magnitude of approximately 6° for flexion-extension, 4° for rotation, and around 10° for lateral flexion.

The intra-rater reliability results are shown in Table 4. The intra-rater analysis for each VR setup yielded ICC values ranging from 0.81 to 0.96 for HTC Vive Pro Eye, from 0.89 to 0.94 for HP Reverb G2, and from 0.92 to 0.93 for Meta Quest 2. The SEM for the flexion-extension ranged from 6° to 8°. For lateral flexion, the mean value was around 5°, and for rotation, there was greater variation with values between 5° and 9°. Regarding the SDD for the flexion-extension movement, it ranged between 18° and 22°,

while for lateral flexion, it ranged between 10° and 15°, and for rotation, between 13° and 25°.

For the usability of the RV setups as CROM assessment instruments, we obtained an overall score of 84 ± 12 for HTC, 86 ± 13 for Meta Quest 2, and 83 ± 15 for HP. There were no reports from the subjects of any adverse events associated with the VR experience.

In addition to quantitative usability scores, informal qualitative feedback was gathered from participants. Overall, the VR setups were perceived as intuitive and easy to follow. However, some participants mentioned discomfort related to headset weight and fit. While this feedback was not analyzed through a formal thematic approach, it provides complementary insights into usability and highlights

areas for refinement in future studies. Additionally, it is worth noting that one participant was unable to complete the assessment with the HP Reverb G2 due to motion sickness symptoms. Although this case was excluded from the analysis, it highlights the importance of considering user tolerance and comfort when evaluating VR-based rehabilitation tools.

4 Discussion

The scientific literature shows considerable variability in the values of CROM for normal populations. This variability is due to the influence of demographic characteristics, such as age and gender, in the studied populations. Consequently, comparing results from different studies that involve distinct populations can be challenging. Despite the variability reported in the literature, our measurements, considering the standard deviation (see Table 3), are consistent with previously published research (Williams et al. 2012; Swinkels and Swinkels-Meewisse 2014; Niewiadomski et al. 2019; Audette et al. 2010), and with previous studies that have used VR to assess CROM (Kiper et al. 2020; Santos-Paz et al. 2022). Although direction-specific analyses (e.g., flexion vs. extension or right vs. left lateral flexion) may provide additional clinical information regarding potential asymmetries, in the present study we reported total ranges within each plane. This decision was motivated by the autonomous nature of the assessment protocol, in which reliably defining and reproducing a neutral position across different directions is challenging and could introduce variability. The approach adopted here facilitates consistency in unsupervised telerehabilitation settings, and has also been used by other researchers (Sarig-Bahat et al. 2010; Raya et al. 2018). Nonetheless, the absence of direction-specific reliability estimates should be acknowledged as a limitation.

It would be reasonable to expect that using the tracker on the back with the HTC Vive Pro Eye to record and correct the user's compensatory movements would lead to more reliable measurements. However, Table 4 reveals a different story. ICC values obtained with the correction are lower for flexion-extension (0.86 vs. 0.81) and remain unchanged for rotation (0.93 in both cases). Only lateral flexion shows a minor improvement (0.94 vs. 0.96). This finding confirms over a new patient population our previous study (Santos-Paz et al. 2022): back movement correction hinders the reliability of flexion-extension, has no significant impact on rotation, and offers a small improvement for lateral rotation.

The rationale behind using the tracker was to subtract compensatory trunk movements from neck movements by subtracting the tracker measurements from the HMD measurements. However, both HMD and tracker measurements

inherently contain errors. These errors stem from limitations in the VR tracking accuracy (Sansone et al. 2022) and from any procedural or user error. We believe that the dominant error source in our tests is the second one; more specifically, the slippage of the HMD and tracker relative to the subject's body. Subtracting HMD measurements from tracker measurements does not cancel out the measurement errors; instead, it adds the measurement errors of both devices due to error propagation (Taylor 1982). Consequently, using the tracker for compensatory movement correction only improves measurement reliability if the subject's compensatory movement is larger than the combined measurement errors of both devices.

This hypothesis is consistent with the study results. The flexo-extension movement involves the most extensive movement, where the user's head reaches a position nearly perpendicular to gravity, then passes through a position parallel to gravity, and finally reaches in the opposite direction another near-perpendicular position. This long motion in which the gravity vector is continually changing with respect to the HMD makes the HMD more susceptible to slippage. Additionally, this movement stretches the back, potentially causing the tracker to shift position. This explains why the subtraction of the trunk movement worsens the ICC (see Table 4) in this case, as the measurement error is large.

The observed decrease in reliability for flexion-extension measurements following tracker correction suggests that additional factors might be influencing the results. One possible explanation is that the tracker may not have been securely fixed in place, leading to unintended movement or slippage during the assessment. A wider and tighter strap may help to more effectively secure the tracker to the back, potentially improving measurement consistency. The posture of the participant could also contribute to variability; if the tracker moves or shifts due to poor alignment or muscle tension, it may result in less accurate readings. By optimizing the tracker attachment and considering these factors, the reliability of CROM measurements may be improved.

In contrast, the rotation movement occurs completely in a plane perpendicular to gravity, experiencing a constant gravitational force on the HMD, likely resulting in less slippage compared to the previous case. This movement also has a lesser effect on the spine and, therefore, the tracker. The only movement where the ICC improves is lateral flexion. Here we expect subjects to exhibit a larger compensatory movement, as this movement often feels restrictive and can be easily compensated for by bending the ipsilateral trunk laterally. By optimizing the tracker attachment and considering these factors, the reliability of CROM measurements may be improved.

The tracker correction showed a variable impact across the different movements. This variation can be attributed to how restrictive each movement is. Lateral flexion, which has the smallest range of motion ($91.75^\circ \pm 14.96^\circ$), showed the highest compensatory movement ($9.56^\circ \pm 9.41^\circ$), corresponding to approximately 10% of the total movement. In comparison, compensatory movements for flexion-extension ($6.36^\circ \pm 6.90^\circ$) and rotation ($3.96^\circ \pm 3.2^\circ$) were lower, representing 5.6% and 2.8% of their respective ranges of motion. This confirms that the greater compensatory movement observed in lateral flexion is what makes the tracker correction more effective, ultimately improving the reliability of this particular measurement.

Given that tracker correction offers minimal to no improvement, and even worsens the reliability of one of the three movements, and considering the added complexity it introduces to the assessment process, the remainder of this section will focus on the measurements from the high-end device without the back tracker correction.

All the CROM mean values for the HTC Vive Pro Eye are slightly smaller than the other two setups (see Table 3), although this difference is not statistically significant (p-value of more than 0.48 for all the movements). The anthropometric information of the subjects who used each setup is similar, with no statistically significant differences between the populations (see Table 2). This effect might be caused by the difference in size of the HMDs, since the HTC Vive Pro Eye is considerably bulkier and heavier (800 g) than the other two (550 and 503 g). The bigger size might make the subject feel that the HMD is going to touch their body and make them stop their movement in that position, instead of achieving their full range of motion; and the heavier weight might impose additional strain on the neck.

Regarding the intra-rater reliability study, good to excellent ICC mean values (Trevethan 2017) were obtained across all devices: HTC Vive Pro Eye, Meta Quest 2, and HP Reverb G2 (see Table 4). Particularly noteworthy were the ICC values for lateral flexion, exceeding 0.90 across all devices, indicating high agreement between measurements for this specific movement. However, it is important to note some differences between devices. While ICC values for lateral flexion were consistently high across all devices, there was variability in ICC values for rotation. In particular, the HP Reverb G2 device showed slightly lower ICC values for rotation compared to the other devices. This might be produced by the looser fitting of this HMD compared to the other two, making the device wobble more at the end of a rotation movement.

The HTC Vive Pro Eye, considered high-end for this study, showed very similar ICC values to the mid-range and low-end setups. This result is interesting because the Light-house tracking system (with two external base stations)

should provide better tracking accuracy than inside-out solutions. However, as previously discussed, we believe that measurement errors in this study were primarily attributed to procedural or user-related factors, such as HMD slippage. While familiarity with the devices may reduce these errors, the influence of headset weight and mass distribution should also be considered. Previous research has shown that heavier HMDs can alter cervical spine torques, potentially affecting measurement consistency (Astrologo et al. 2024). Despite no visible slippage during tests, the increased weight of the HTC Vive Pro Eye likely leads to greater variability in measurements due to increased torque and potential relative movements between the HMD and the head.

In this regard, the bulkier nature of the HTC Vive Pro Eye compared to the other two options seems to offset the benefits of using a superior tracking system. Although high-end setups generally provide superior tracking accuracy, their increased weight might introduce additional variability in repeated CROM measurements. Future studies should further investigate the impact of HMD weight on reliability, particularly in extended use scenarios. The difference in movement tracking frequencies between devices (90 Hz in high-end and mid-range setups, and 50 Hz in low-end setups) had a negligible impact on the results.

One limitation of this study is that only one evaluator was involved in the assessments, which ensures consistency in intra-rater reliability but limits the generalizability of the findings. Future studies should incorporate multiple evaluators to assess inter-rater reliability. However, it is worth noting that a previous CROM assessment study (Santos-Paz et al. 2022) using a high-end VR setup and two independent raters found high inter-rater reliability (ICC = 0.83–0.97), comparable to the intra-rater reliability values (ICC = 0.86–0.96). This suggests that the method is robust across different evaluators.

The test–retest measurements of the study were conducted on the same day, which may have led to an overestimation of reliability due to memory effects, consistent positioning, and reduced procedural variability. To partially mitigate order and learning effects, the movements within each assessment were presented in a randomized order for each participant; however, this approach does not fully eliminate the influence of motor memory or stable conditions between sessions. Future studies should consider a delayed retest (e.g., one week apart) to better assess temporal consistency of measurements in real-world rehabilitation settings. Another limitation of the study is the lack of a formal a priori power analysis; the sample size was only based on the minimum group sizes recommended for ICC studies (Koo and Li 2016).

Regarding the usability of VR devices as CROM assessment instruments, we observed consistent scores across

different devices. Although Meta Quest 2 had the highest score, the differences were so small as to be negligible. And, since the three devices run the same application, any difference in score can only be attributed to each device's physical characteristics. In addition to the SUS scores, informal feedback from participants was also collected regarding the comfort and usability of the HMDs using a free text field in the form. A few participants, particularly female, commented that the HTC Vive Pro Eye felt a little uncomfortable, but without reaching the point of affecting their movements. With the HP Reverb G2, a single participant reported experiencing dizziness during a measurement, which led to the discontinuation of the trial. This measurement was discarded, as it was not completed.

Device stability plays a crucial role in ensuring reliable measurements during CROM assessments. The HTC Vive Pro Eye features rigid head straps, providing a more stable fit. However, its heavier weight (800 g) may lead to slippage if not properly adjusted. On the other hand, the Meta Quest 2 uses elastic straps, which may make adjustment slightly more challenging, but also lead to tighter fits. This is the only HMD studied that is not tethered to a computer by a cable, thus eliminating any potential impact of cable weight or resistance on movement. Finally, this HMD is lighter (503 g), reducing the impact on slippage. The HP Reverb G2 (550 g) also features rigid head straps, but they have a worse fit than those of the HTC Vive Pro Eye. Adjustments to the fit and overall design could help reduce the movement of the headset relative to the head, thereby minimizing variability in the data. A tighter fit, achieved through more adjustable straps or the use of wider, padded bands, could improve stability without sacrificing user comfort. Additionally, reducing the weight or redistributing the center of mass to better align with the head's natural balance could further enhance stability (Astrologo et al. 2024). These adjustments could not only improve measurement reliability but also reduce the likelihood of discomfort or slippage during testing, particularly in dynamic movements.

The recommendation to prioritize low-cost setups is based on their accessibility and ease of deployment, particularly in clinical and at-home rehabilitation contexts. However, there are scenarios where high-end setups may offer advantages that justify their higher cost and complexity. High-end devices, such as the HTC Vive Pro Eye, provide superior immersion through better graphics, which could enhance patient engagement and adherence to long-term rehabilitation programs. Furthermore, it is important to note that inside-out tracking has limitations, particularly in detecting hands or controllers when they are out of sight of the device's cameras, which may occur more frequently when patients perform assessments without direct supervision.

Nevertheless, these benefits must be weighed against practical considerations. High-end setups require more complex calibration, dedicated space, and may not be feasible for widespread at-home use. In contrast, low-cost setups with inside-out tracking, like Meta Quest 2, are simpler to set up and more practical for remote monitoring. Ultimately, the choice of device should consider the specific usage scenario.

Seven out of 85 participants ($\approx 8\%$) used more than one VR setup, while most participants were tested under a single condition. Although this uneven exposure was rare and is unlikely to have substantially biased results, we acknowledge that it could introduce minor carryover effects and limit strict between-group comparability.

5 Conclusion

This study evaluated the reliability of three VR setups (HTC Vive Pro Eye, HP Reverb G2, and Meta Quest 2) as CROM assessment tools. These devices represent three distinct categories of VR setups: high-end systems with external base station tracking and a PC-tethered HMD, mid-range systems with inside-out tracking and a PC-tethered HMD, and low-end systems with inside-out tracking and a stand-alone HMD. At least 30 healthy subjects were assessed twice by the same rater using each VR setup (see Table 2). The CROM values (see Table 3) indicate high consistency across all devices, and the mean ICC values (see Table 4) show good to excellent reliability, supporting their use as CROM assessment instruments. Moreover, the usability scores (SUS) were similar across all devices, suggesting an effective user experience for CROM evaluation independent of the device.

Our findings indicate that low-cost VR setups offer comparable reliability to higher-end systems for CROM assessment, challenging the need for expensive equipment in clinical settings. Factors such as the physical design, weight distribution, and gripping mechanism of the HMD had a greater influence on measurement consistency than the theoretical accuracy of the tracking system. These results suggest the potential for developing telerehabilitation solutions using low-cost VR setups, increasing accessibility and reducing costs in rehabilitation. While our results suggest that low-cost devices can provide sufficient reliability for certain applications, the choice of equipment should consider the specific requirements of each clinical or research setting.

A hybrid model, combining in-person clinical sessions with remote VR-guided exercises, could allow patients to engage in therapy from home while clinicians monitor their progress through session metrics. This approach could

reduce the frequency of in-person visits, ease healthcare system burdens, shorten waiting times, and lower costs without compromising care quality. However, certain groups, particularly the elderly, may hesitate to adopt these technologies and prefer in-person rehabilitation. Even so, partial adoption could still alleviate strain on healthcare workers, decrease costs, and reduce waitlists, as some patients may opt for this solution.

Based on the results of this study, low-cost VR setups demonstrated no statistically significant differences in reliability when compared to high-end systems for CROM assessment. However, these findings should be interpreted within the context of the study's limitations. Our analysis focused on intra-rater reliability and examined repeatability rather than accuracy. Additionally, factors such as HMD weight and mass distribution, which may influence measurement consistency, were not systematically analyzed. Future studies should explore the influence of different evaluators, accuracy validation against a ground-truth system, and the long-term usability of various VR setups in real-world conditions.

Next steps for validating VR-based CROM assessments include studies with clinical populations, particularly patients experiencing cervical pain, to evaluate the reliability and effectiveness of these methods in therapeutic contexts. The HTC Vive Pro Eye, with its external tracking, may prove beneficial for these patients due to potential compensatory movements resulting from pain. It will also be crucial to assess the feasibility of using these systems in patients' homes, considering factors such as environment, posture, and adherence to instructions, which may affect the accuracy of the assessments.

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Author contributions J.S. was the main developer of the virtual reality experience, participated in data collection and analysis, and wrote the first draft of the manuscript. R.G. collaborated in the conceptualization and methodological design of the experiments, software validation, interpretation of the results, and manuscript writing. A.S. collaborated in the development of the virtual reality experience and participated in data collection and analysis. A.O. collaborated in the conceptualization and methodological design of the experiments, interpretation of the results, and manuscript writing. All authors reviewed the manuscript.

Data availability All the data and materials in the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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References

- Afonso J, Brito J, Abade E, Rendeiro-Pinho G, Baptista I, Figueiredo P, Nakamura FY (2024) Revisiting the 'whys' and 'hows' of the warm-up: Are we asking the right questions? *Sports Med* 54(1):23–30. <https://doi.org/10.1007/s40279-023-01908-y>
- Audette I, Dumas J-P, Côté JN, De Serres SJ (2010) Validity and between-day reliability of the cervical range of motion (CROM) device. *J Orthop Sports Phys Therapy* 40(5):318–323. <https://doi.org/10.2519/jospt.2010.3180>
- Astrologo AN, Nano S, Klemm EM, Shefelbine SJ, Dennerlein JT (2024) Determining the effects of AR/VR HMD design parameters (mass and inertia) on cervical spine joint torques. *Appl Ergon* 116:104183. <https://doi.org/10.1016/j.apergo.2023.104183>
- Bhandary M, Fujiwara K (2013) Test for intraclass correlation coefficient under unequal family sizes. *J Mod Appl Stat Methods* 12(2):171–183. <https://doi.org/10.22237/jmasm/1383278880>
- Bangor A, Kortum PT, Miller JT (2008) An empirical evaluation of the system usability scale. *Intl J Human-Comput Interact* 24(6):574–594. <https://doi.org/10.1080/10447310802205776>
- Brepohl PCA, Leite H (2023) Virtual reality applied to physiotherapy: a review of current knowledge. *Virtual Real* 27(1):71–95. <https://doi.org/10.1007/s10055-022-00654-2>
- Brooke J (1996) Sus: a 'quick and dirty' usability scale. In: Jordan, P.W, Thomas, B, Weerdmeester, B.A, McClelland, I.L. (eds.) *Usability evaluation in industry*, 1st edn., pp. 189–194. CRC Press, London. <https://doi.org/10.1201/9780429157011-35>
- Cánovas-Ambit G, García-Vidal JA, Martín-San Agustín R, Dalla-Vecchia AA, Sánchez-Barbadora M, Medina-Mirapeix F (2021) Validity and reliability of Veloflex to measure active cervical range of motion in asymptomatic and symptomatic subjects. *PeerJ* 9:11228. <https://doi.org/10.7717/peerj.11228>
- Cavedoni S, Cipresso P, Mancuso V, Bruni F, Pedrolì E (2022) Virtual reality for the assessment and rehabilitation of neglect: Where are we now? a 6-year review update. *Virtual Real* 26(4):1663–1704. <https://doi.org/10.1007/s10055-022-00648-0>
- Câmara J, Ferreira R, Teixeira L, Nóbrega J, Romeira C, Badia SBI, Faria AL (2023) Efficacy of adaptive cognitive training through desktop virtual reality and paper-and-pencil in the treatment of mental and behavioral disorders. *Virtual Real* 27(1):291–306. <https://doi.org/10.1007/s10055-021-00559-6>
- Cerda IH, Therond A, Moreau S, Studer K, Donjow AR, Crowther JE, Mazzolenis ME, Lang M, Tolba R, Gilligan C et al (2024) Telehealth and virtual reality technologies in chronic pain management: a narrative review. *Curr Pain Headache Rep* 28(3):83–94. <https://doi.org/10.1007/s11916-023-01205-3>
- Chehade MJ, Yadav L, Kopansky-Giles D, Merolli M, Palmer E, Jayatilaka A, Slater H (2020) Innovations to improve access to musculoskeletal care. *Best Pr Res Clin Rheumatol* 34(5):101559. <https://doi.org/10.1016/j.berh.2020.101559>

- González-González CS, Toledo-Delgado PA, Muñoz-Cruz V, Torres-Carrion PV (2019) Serious games for rehabilitation: gestural interaction in personalized gamified exercises through a recommender system. *J Biomed Inf* 97:103266. <https://doi.org/10.1016/j.jbi.2019.103266>
- Gumaa M, Khaireldin A, Rehan Youssef A (2021) Validity and reliability of interactive virtual reality in assessing the musculoskeletal system: a systematic review. *Curr Rev Musculoskelet Med* 14:130–144. <https://doi.org/10.1007/s12178-021-09696-6>
- Huang C-Y, Chiang W-C, Yeh Y-C, Fan S-C, Yang W-H, Kuo H-C, Li P-C (2022) Effects of virtual reality-based motor control training on inflammation, oxidative stress, neuroplasticity and upper limb motor function in patients with chronic stroke: a randomized controlled trial. *BMC Neurol* 22(1):21. <https://doi.org/10.1186/s12883-021-02547-4>
- Janjua MB, Zhou PL, Vasquez-Montes D, Moskovich R (2020) Photogrammetric analysis: an objective measure to assess the craniocervical range of motion after cervical laminoplasty surgeries. *J Clin Neurosci* 71:76–83. <https://doi.org/10.1016/j.jocn.2019.10.012>
- Knols RH, Aufdemkampe G, De Bruin ED, Uebelhart D, Aaronson NK (2009) Hand-held dynamometry in patients with haematological malignancies: measurement error in the clinical assessment of knee extension strength. *BMC Musculoskelet Disord* 10(1):31. <https://doi.org/10.1186/1471-2474-10-31>
- Kiper P, Baba A, Alhelou M, Pregnotato G, Maistrello L, Agostini M, Turolla A (2020) Assessment of the cervical spine mobility by immersive and non-immersive virtual reality. *J Electromyogr Kinesiol* 51:102397. <https://doi.org/10.1016/j.jelekin.2020.102397>
- Koo TK, Li MY (2016) A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med* 15(2):155–163. <https://doi.org/10.1016/j.jcm.2016.02.012>
- Kazeminasab S, Nejadghaderi SA, Amiri P, Pourfathi H, Araj-Khodaei M, Sullman MJ, Kolahi A-A, Safiri S (2022) Neck pain: global epidemiology, trends and risk factors. *BMC Musculoskelet Disord* 23:1–13. <https://doi.org/10.1186/s12891-021-04957-4>
- Knudson DV (2018) Warm-up and flexibility. In: Chandler TJ, Brown LE (eds) *Conditioning for strength and human performance*, 3rd edn., pp 212–231. Routledge, Philadelphia, PA. <https://doi.org/10.4324/9781315438450-11>
- Kauther MD, Piotrowski M, Hussmann B, Lendemans S, Wedemeyer C (2012) Cervical range of motion and strength in 4,293 young male adults with chronic neck pain. *Eur Spine J* 21:1522–1527. <https://doi.org/10.1007/s00586-012-2369-x>
- Kammler-Sücker KI, Löffler A, Flor H (2023) Effects of personalized movement models in virtual reality on pain expectancy and motor behavior in patients with chronic back pain: a feasibility study. *Virtual Real* 27(4):3581–3598. <https://doi.org/10.1007/s10055-023-00800-4>
- Lewis R, Gómez Álvarez CB, Rayman M, Lanham-New S, Woolf A, Mobasher A (2019) Strategies for optimising musculoskeletal health in the 21st century. *BMC Musculoskelet Disord* 20:1–15. <https://doi.org/10.1186/s12891-019-2510-7>
- Luedtke K, Schoettler-Königer T, Hall T, Reimer C, Grassold M, Haselhoff-Styler P, Neulinger C, Obrocki M, Przyhoda P, Schäfer A (2020) Concurrent validity and reliability of measuring range of motion during the cervical flexion rotation test with a novel digital goniometer. *BMC Musculoskelet Disord* 21:1–10. <https://doi.org/10.1186/s12891-020-03525-6>
- Moulai K, Sheikhtaheri A, Nezhad MS, Haghdost A, Gheysari M, Bahaadinbeigi K (2022) Telerehabilitation for upper limb disabilities: a scoping review on functions, outcomes, and evaluation methods. *Arch Publ Health* 80(1):196. <https://doi.org/10.1186/s13690-022-00952-w>
- Niewiadomski C, Bianco R-J, Afquir S, Evin M, Arnoux P-J (2019) Experimental assessment of cervical ranges of motion and compensatory strategies. *Chiropr Man Ther* 27:1–9. <https://doi.org/10.1186/s12998-018-0223-x>
- Niehorster DC, Li L, Lappe M (2017) The accuracy and precision of position and orientation tracking in the HTC vive virtual reality system for scientific research. *i-Perception* 8(3), 2041669517708205 <https://doi.org/10.1177/2041669517708205>
- Prushansky T, Deryi O, Jabbar B (2010) Reproducibility and validity of digital inclinometry for measuring cervical range of motion in normal subjects. *Physiother Res Int* 15(1):42–48. <https://doi.org/10.1002/pri.443>
- Rojo A, Cortina J, Sánchez C, Urendes E, García-Carmona R, Raya R (2022) Accuracy study of the oculus touch v2 versus inertial sensor for a single-axis rotation simulating the elbow's range of motion. *Virtual Real* 26(4):1651–1662. <https://doi.org/10.1007/s10055-022-00660-4>
- Raya R, García-Carmona R, Sanchez C, Urendes E, Ramirez O, Martín A, Otero A (2018) An inexpensive and easy to use cervical range of motion measurement solution using inertial sensors. *Sensors* 18(8):2582. <https://doi.org/10.3390/s18082582>
- Sarig-Bahat H, Weiss PLT, Laufer Y (2010) Neck pain assessment in a virtual environment. *Spine* 35(4):105–112. <https://doi.org/10.1097/BRS.0b013e3181b79358>
- Snodgrass SJ, Cleland JA, Haskins R, Rivett DA (2014) The clinical utility of cervical range of motion in diagnosis, prognosis, and evaluating the effects of manipulation: a systematic review. *Physiotherapy* 100(4):290–304. <https://doi.org/10.1016/j.physio.2014.04.007>
- Sevilla-Gonzalez MDR, Loaeza LM, Lazaro-Carrera LS, Ramirez BB, Rodríguez AV, Peralta-Pedrero ML, Almeda-Valdes P (2020) Spanish version of the system usability scale for the assessment of electronic tools: development and validation. *JMIR Hum Factors* 7(4):21161. <https://doi.org/10.2196/21161>
- Santos-Paz JA, Sánchez-Picot Á, Rojo A, Martín-Pintado-Zugasti A, Otero A, García-Carmona R (2022) A novel virtual reality application for autonomous assessment of cervical range of motion: development and reliability study. *PeerJ* 10:14031. <https://doi.org/10.7717/peerj.14031>
- Sansone LG, Stanzani R, Job M, Battista S, Signori A, Testa M (2022) Robustness and static-positional accuracy of the steamvr 1.0 virtual reality tracking system. *Virtual Real* 26(3):903–924. <https://doi.org/10.1007/s10055-021-00584-5>
- Swinkels RA, Swinkels-Meewisse IE (2014) Normal values for cervical range of motion. *Spine* 39(5):362–367. <https://doi.org/10.1097/BRS.0000000000000158>
- Stoustrup P, Verna C, Kristensen KD, Küsel A, Herlin T, Pedersen TK (2013) Smallest detectable differences in clinical functional temporomandibular joint examination variables in juvenile idiopathic arthritis. *Orthod Craniofac Res* 16(3):137–145. <https://doi.org/10.1111/ocr.12008>
- Taylor JR (1982) *An introduction to error analysis: the study of uncertainties in physical measurements*. University Science Books, Mill Valley, CA, p 270
- Trevethan R (2017) Intraclass correlation coefficients: clearing the air, extending some cautions, and making some requests. *Health Serv Outcomes Res Method* 17(2):127–143. <https://doi.org/10.1007/s10742-016-0156-6>
- Tsang SM, Szeto GP, Lee RY (2013) Movement coordination and differential kinematics of the cervical and thoracic spines in people with chronic neck pain. *Clin Biomech* 28(6):610–617. <https://doi.org/10.1016/j.clinbiomech.2013.05.009>
- Kruk E, Silverman AK, Koizia L, Reilly P, Fertleman M, Bull AM (2021) Age-related compensation: neuromusculoskeletal capacity, reserve & movement objectives. *J Biomech* 122:110385. <https://doi.org/10.1016/j.jbiomech.2021.110385>

- Williams MA, McCarthy CJ, Chorti A, Cooke MW, Gates S (2010) A systematic review of reliability and validity studies of methods for measuring active and passive cervical range of motion. *J Manipulative Physiol Ther* 33(2):138–155. <https://doi.org/10.1016/j.jmpt.2009.12.009>
- Wolan-Nieroda A, Guzik A, Mocur P, Druzbicki M, Maciejczak A et al (2020) Assessment of interrater and intrarater reliability of cervical range of motion (crom) goniometer. *BioMed Res Int* **2020** <https://doi.org/10.1155/2020/8908035>
- World Health Organization (2022) Musculoskeletal conditions. <https://www.who.int/news-room/fact-sheets/detail/musculoskeletal-conditions>. Accessed: March 22, 2024
- Williams MA, Williamson E, Gates S, Cooke MW (2012) Reproducibility of the cervical range of motion (CROM) device for individuals with sub-acute whiplash associated disorders. *Eur Spine J* 21:872–878. <https://doi.org/10.1007/s00586-011-2096-8>
- Xu X, Chen KB, Lin J-H, Radwin RG (2015) The accuracy of the Oculus rift virtual reality head-mounted display during cervical spine mobility measurement. *J Biomech* 48(4):721–724. <https://doi.org/10.1016/j.jbiomech.2015.01.005>
- Yoon T-L, Kim H-N, Min J-H (2019) Validity and reliability of an inertial measurement unit-based 3-dimensional angular measurement of cervical range of motion. *J Manipulative Physiol Ther* 42(1):75–81. <https://doi.org/10.1016/j.jmpt.2018.06.001>
- Yoo W-G, Park S-Y, Lee M-R (2011) Relationship between active cervical range of motion and flexion-relaxation ratio in asymptomatic computer workers. *J Physiol Anthropol* 30(5):203–207. <https://doi.org/10.2114/jpa2.30.203>

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